Table 3 Overview of included studies on health statue utility values

Region	Study Author year	Study design/study type	Patient population	Treatment	Follow-up times assessed	Utility assessment method	Scale	Baseline mean (SD)	Mean change at last follow-up (SD)
Utilities									
EQ-5D									
Global	CheckMate 436 Zinzani 2019 [45]	Phase 1/2 single-arm study	Adult patients with R/R PMBCL	Nivolumab + brentuximab vedotin	Baseline Cycle 5 Follow-up 1	EQ-5D-3L mobility	0 (no problems) to 3 (extreme problems)	Some problems: 14% Extreme problems: 0	Some problems: 21% Extreme problems: 0
						EQ-5D-3L self-care	0 (no problems) to 3 (extreme problems)	Some problems: 11% Extreme problems: 0	Some problems: 14% Extreme problems: 0
						EQ-5D-3L activity	0 (no problems) to 3 (extreme problems)	Some problems: 39% Extreme problems: 0	Some problems: 21% Extreme problems: 0
						EQ-5D-3L pain	0 (no problems) to 3 (extreme problems)	Some problems: 71% Extreme problems: 7%	Some problems: 21% Extreme problems: 7%
						EQ-5D-3L anxiety	0 (no problems) to 3 (extreme problems)	Some problems: 46% Extreme problems: 0	Some problems: 57% Extreme problems: 0
Global	SADAL Shah 2021 [38] Casasnovas 2020 [69]	Phase 2b single-arm trial	Adult patients with R/R DLBCL with 2 + prior lines of therapy	Selinexor	Baseline Cycle 2 Cycle 3 Cycle 4 Cycle 5 Cycle 6 Cycle 7	EQ-5D-5L health index score	0 (dead) to 1 (perfect health)	Responders: 0.789 (95% CI, 0.767, 0.811) Nonresponders: 0.801 (95% CI, 0.781, 0.822)	Responders: − 0.010 (95% CI, − 0.15, − 0.05) Nonresponders: − 0.274 (NR)
Global	ZUMA-1 Lin 2019 [42] ClinicalTrials.gov [70]	Phase 2 single-arm trial	Adult patients with 3L + DLBCL of therapy (Cohorts 3−6)	Axi-cel	Baseline Week 4 Month 3 Month 6	EQ-5D-5L visual analog scale	0 to 100 (higher scores reflecting better health)	Cohort 3: 71.2 (21.3) Cohort 4: 69.5 (18.8) Cohort 5: 66.7 (20.7) Cohort 6: 70.9 (17.0)	Cohort 3: + 5.9[a] Cohort 4: + 15.6[a] Cohort 5: + 10.4[a] Cohort 6: + 8.9[a]
			Adult patients with 3L + LBCL (safety management study)		Baseline 1 month 2 months 3 months 6 months	EQ-5D-5L health index score	0 (dead) to 1 (perfect health)	0.80 (0.17)	0.82 (0.21)
US	TRANSCEND NHL 001 Patrick 2021 [17] Patrick 2020 [37] Patrick 2019 [35] Patrick 2019 [36]	Phase 1 single-arm trial	Adult patients with R/R DLBCL with 2 + prior lines of therapy	Liso-cel	Baseline 1 month 2 months 3 months 6 months 9 months 12 months 18 months	EQ-5D-5L health index score	0 (dead) to 1 (perfect health)[a]	0.817 (0.120)	+ 0.061[b] (0.013)
						EQ-5D-5L visual analog scale	0 to 100 (higher scores reflecting better health)	68.3 (19.5)	+ 10.4 (5.4)
NR	Orfanos 2022 [46]	Phase 2 single-arm trial	2L and heavily pretreated patients with R/R DLBCL	Naratuximab emtansine + rituximab	Baseline End of therapy	EQ-5D (unspecified)	0 (dead) to 1 (perfect health)	Responders: 0.78 (NR) Nonresponders: 0.73 (NR)	Responders: 0.77 (NR) Nonresponders: 0.67 (NR)

Table 3 (continued)

Region	Study Author year	Study design/study type	Patient population	Treatment	Follow-up times assessed	Utility assessment method	Scale	Baseline mean (SD)	Mean change at last follow-up (SD)
UK	Howell 2022 [44] Howell 2020 [43]	Vignette-based study	General adult population in UK	Hypothetical CAR T cell therapy pathway	NA	EQ-5D-5L health index score	0 (dead) to 1 (perfect health)[c]	0.86 (0.17)	NR
						EQ-5D-5L visual analog scale	0 to 100 (higher scores reflecting better health)	82.15 (13.54)	NR
TTO									
UK	Howell 2022 [44] Howell 2020 [43]	Vignette-based study	General adult population in UK	Hypothetical CAR T cell therapy pathway	NA	TTO (1-year time horizon)	0 (dead) to 1 (perfect health)[c]	NR (results are disutilities of a hypothetical treatment pathway [i.e., no baseline])	CAR T cell therapy for LBCL with grade 1 CRS: − 0.01 (0.04)[d] CAR T cell therapy for LBCL with grade 2 CRS: − 0.05 (0.09)[d] CAR T cell therapy for LBCL with grade 3/4 CRS: − 0.23 (0.24)[d] CAR T cell therapy for LBCL with grade 1/2 NEs: − 0.04 (0.07)[d] CAR T cell therapy with grade 3/4 NEs: − 0.18 (0.22)[d]

2L second line; *3L+* third line or later; *AE* adverse event; *axi-cel* axicabtagene ciloleucel; *CAR* chimeric antigen receptor; *CI* confidence interval; *CRS* cytokine release syndrome; *DLBCL* diffuse large B cell lymphoma; *EQ-5D-3L* EuroQol 5-dimensional, 3-level; *EQ-5D-5L* EuroQol 5-dimensional 5-level; *LBCL* large B cell lymphoma; *liso-cel* lisocabtagene maraleucel; *NA* not available; *NE* neurological event; *NR* not reported; *PMBCL* primary mediastinal large B cell lymphoma; *R/R* relapsed or refractory; *SD* standard deviation; *TTO* time tradeoff; *UK* United Kingdom; *US* United States

[a]EQ-5D visual analog scale values are calculated using the 6-month end score

[b]Mean change was calculated based on baseline and follow-up values

[c]Utility values can occasionally be below zero, reflecting a health state less preferable than dead

[d]Values represent the difference between the utility of the health state without an AE (i.e., base LBCL health state) and with an AE

Table 4 Overview of economic studies reporting health state utility values

Author, year	Region	Study type	Patient population	Treatment	Utility values on-treatment	Utility values in remission	Utility values for disease progression
Li 2022 [54]	China	CUA	Adult patients with R/R DLBCL with ≥ 2 prior lines of systemic therapies	Axi-cel Salvage chemotherapy	Disutility of chemotherapy: − 0.42[a]	0.83	0.39
Wakase 2021 [55]	Japan	CUA	Adult patients with CD19-positive R/R DLBCL who are ineligible for, or relapsed after, autologous HSCT	Tisagenlecleucel Salvage chemotherapy	− 0.15; 28 days[b] − 0.15; 62 days[b]	0.83	0.39
Cher 2020 [53]	Singapore	CUA	Adult patients with R/R DLBCL from JULIET trial and CORAL extension study	Tisagenlecleucel Salvage chemotherapy	Disutility: − 0.15[c] Disutility: − 0.15[c]	0.7	0.59
Wang 2021 [58]	Singapore	CUA	Adult patients with R/R DLBCL with ≥ 2 prior lines of systemic therapies	Tisagenlecleucel Salvage chemotherapy	Disutility: − 0.15[d] Disutility: − 0.15[d]	0.90 (SE, 0.01) 0.90 (SE, 0.01)	0.82 (SE, 0.02) 0.82 (SE, 0.02)
Bastos-Oreiro 2022 [71]	Spain	CUA	Adult patients with R/R DLBCL with ≥ 2 prior lines of systemic therapies	Axi-cel Tisagenlecleucel	Month 1: 0.740	≤ 12 months: 0.782 > 12 months: 0.820	0.39
Muszbek 2016 [62]	UK	CUA	Patients with R/R aggressive NHL receiving third- or fourth-line treatment	Pixantrone	Grade 2 AE disutility: 0.0075 Grade 3/4 AE disutility: 0.0078	Preprogression: 0.76 (SE, 0.03)	0.68 (SE, 0.04)
				Current clinical practice	Grade 2 AE disutility: 0.0066 Grade 3/4 AE disutility: 0.0073		
Betts 2020 [61] Betts 2020 [72]	US	CEA	Adult patients with R/R DLBCL based on GO29365 trial	Pola-BR BR	NR	0.83	0.71
Liu 2021 [49]	US	CUA	Adult patients with R/R DLBCL with ≥ 2 prior lines of systemic therapies	Axi-cel Tisagenlecleucel	Month 1: 0.740	First 24 months: 0.782 > 24 months since: 0.820	0.39
Lin 2019 [50]	US	CUA	US adults with R/R DLBCL after ≥ 2 lines of therapy or relapsed ≤ 12 months after HSCT	Axi-cel	Months 1–2: 0.50	0.70	0.45
				Tisagenlecleucel	Months 1–2: 0.58	0.70	
				Chemoimmunotherapy	During treatment: 0.63	0.71	
				Autologous HSCT	Months 1–2: 0.43 Month 3 (if in remission): 0.70	0.70	

Table 4 (continued)

Author, year	Region	Study type	Patient population	Treatment	Utility values on-treatment	Utility values in remission	Utility values for disease progression
Oluwole 2022 [48]	US	CUA	US adults with R/R LBCL after ≥ 2 lines of therapy	Axi-cel Liso-cel	First model cycle: 0.740	≤ 24 months since model entry: 0.782 > 24 months since model entry: 0.820	0.390
Roth 2018 [51]	US	CUA	Adult patients with R/R DLBCL based on ZUMA-1 trial	Axi-cel Salvage chemotherapy	0.740 0.673	< 6-month follow-up: 0.782 ≥ 6-month follow-up: 0.823	0.390
Whittington 2019 [56]	US	CUA	Adult patients with R/R B cell lymphoma from ZUMA-1 trial	Axi-cel Chemotherapy HSCT	NR Disutility: − 0.42 Disutility: − 0.57	0.83	0.39
Calamia 2021 [59]	US	CUA	Adult patients with R/R DLBCL who are transplant ineligible based on L-MIND and G029365 trials	Pola-BR Tafasitamab + lenalidomide	AE disutility: − 0.02 AE disutility: − 0.01	PFS: 0.83 PFS adjusted for disutility: 0.82	0.39
Patel 2020 [60]	US	CUA	Adult patients with R/R follicular or DLBCL (mirrored phase 2 cohort from NCT02257567)	Pola-BR BR	Disutility from chemotherapy: − 0.42	0.83	0.39
Qi 2021 [57]	US	CUA	Adults with R/R DLBCL after ≥ 2 lines of therapy	Tisagenlecleucel Salvage chemotherapy	Disutility: − 0.15 (duration: 27.9 days)[e] Disutility: − 0.15 (duration: 72.2 days)[e]	0.83	0.39

AE adverse event; *axi-cel* axicabtagene ciloleucel; *BR* bendamustine and rituximab; *CRS* cytokine release syndrome; *CEA* cost-effectiveness analysis; *CUA* cost-utility analysis; *DLBCL* diffuse large B cell lymphoma; *HSCT* hematopoietic stem cell transplantation; *ICU* intensive care unit; *NHL* non-Hodgkin lymphoma; *NR* not reported; *PFS* progression-free survival; *pola-BR* polatuzumab vedotin, bendamustine, and rituximab; *R/R* relapsed or refractory; *SE* standard error; *UK* United Kingdom; *US* United States

[a]Reports additional disutility value for HSCT (− 0.57)

[b]Reports additional disutility value for subsequent HSCT (− 0.30/year)

[c]Disutilities were also provided for ICU stay for CRS (− 0.7), AEs (− 0.15), and HSCT (− 0.15)

[d]Disutilities were also provided for ICU stay for CRS (− 0.9), ICU stay for non-CRS AEs (− 0.9), and subsequent HSCT (− 0.30/year)

[e]Disutilities were also provided for ICU stay for CRS (− 0.83; duration 8.5 days) and ICU stay for non-CRS AEs (− 0.83; duration 0.9 days) for tisagenlecleucel-infused patient, and subsequent HSCT (− 0.30/year) for both treatments